

CSE5DMI Data Mining

Week 10
CLUSTER ANALYSIS: BASIC CONCEPTS AND ALGORITHMS

La Trobe University CRICOS Provider Code Number 00115M

Quick review: Week 9

Clustering basics

K-means

DBSCAN

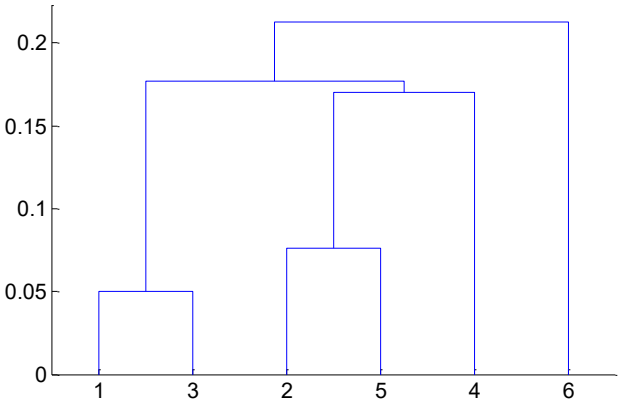
This week

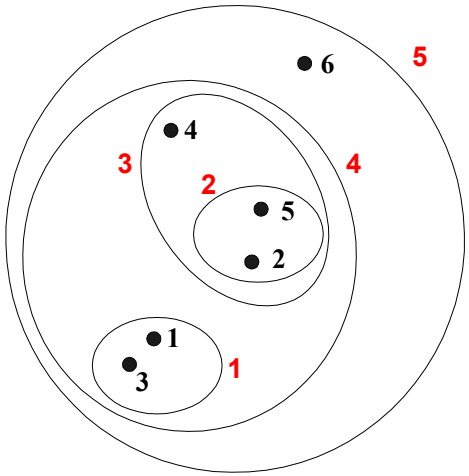
Hierarchical Clustering

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Hierarchical Clustering

- Produces a set of nested clusters organized as a hierarchical tree
- Can be visualized as a dendrogram
  - A tree like diagram that records the sequences of merges or splits





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Strengths of Hierarchical Clustering

- Do not have to assume any particular number of clusters
  - Any desired number of clusters can be obtained by ‘cutting’ the dendrogram at the proper level
- They may correspond to meaningful taxonomies
  - Example in biological sciences (e.g., animal kingdom, phylogeny reconstruction, …)

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Hierarchical Clustering

- Two main types of hierarchical clustering
  - Agglomerative:
    - Start with the points as individual clusters
    - At each step, merge the closest pair of clusters until only one cluster (or k clusters) left
  - Divisive:
    - Start with one, all-inclusive cluster
    - At each step, split a cluster until each cluster contains an individual point (or there are k clusters)
- Traditional hierarchical algorithms use a similarity or distance matrix
  - Merge or split one cluster at a time

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Agglomerative Clustering Algorithm

- **Key Idea: Successively merge closest clusters**
- Basic algorithm

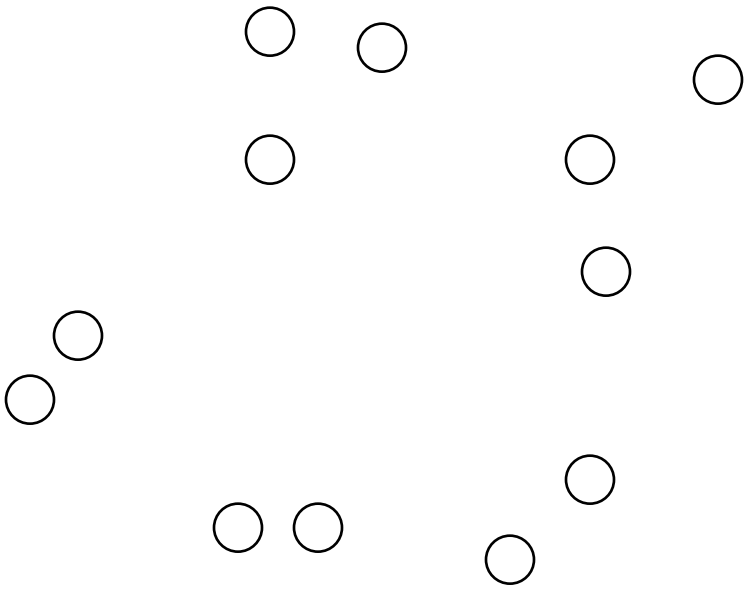
1. Compute the proximity matrix
2. Let each data point be a cluster
3. **Repeat**
4. Merge the two closest clusters
5. Update the proximity matrix
6. **Until** only a single cluster remains

- Key operation is the computation of the proximity of two clusters
  - Different approaches to defining the distance between clusters distinguish the different algorithms

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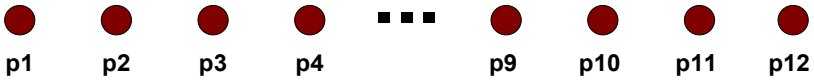
Steps 1 and 2

- Start with clusters of individual points and a proximity matrix



	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

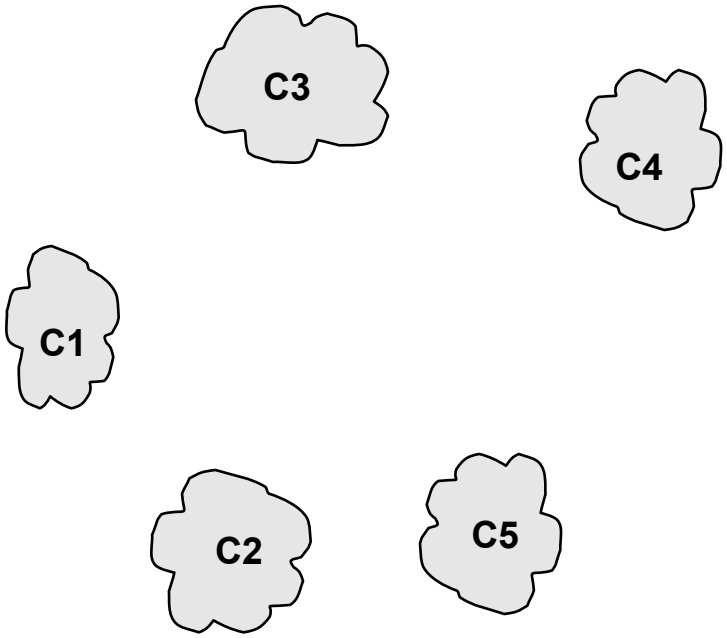
Proximity Matrix



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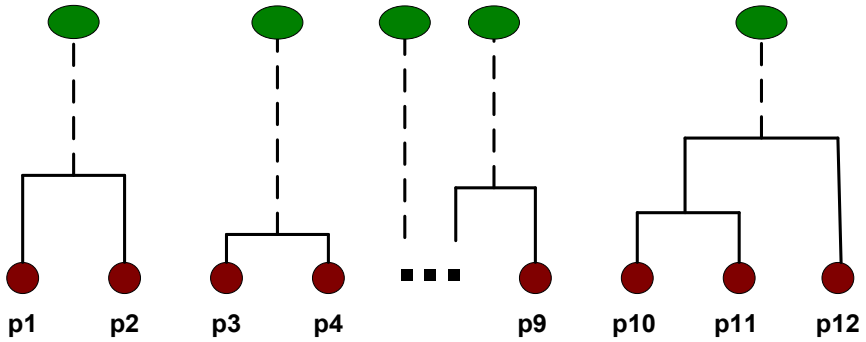
Intermediate Situation

- After some merging steps, we have some clusters



	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

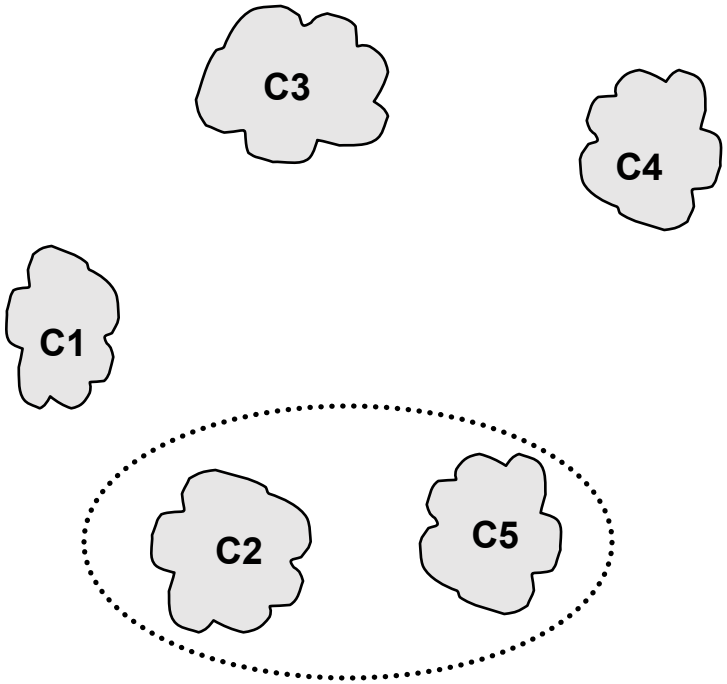
Proximity Matrix



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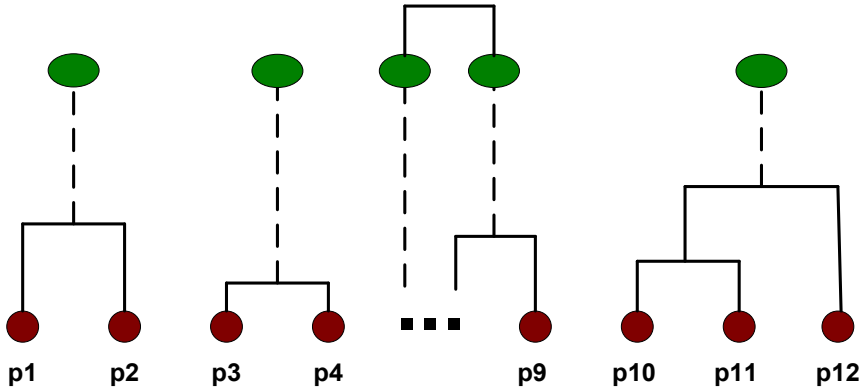
Step 4

- We want to merge the two closest clusters (C2 and C5) and update the proximity matrix.



	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

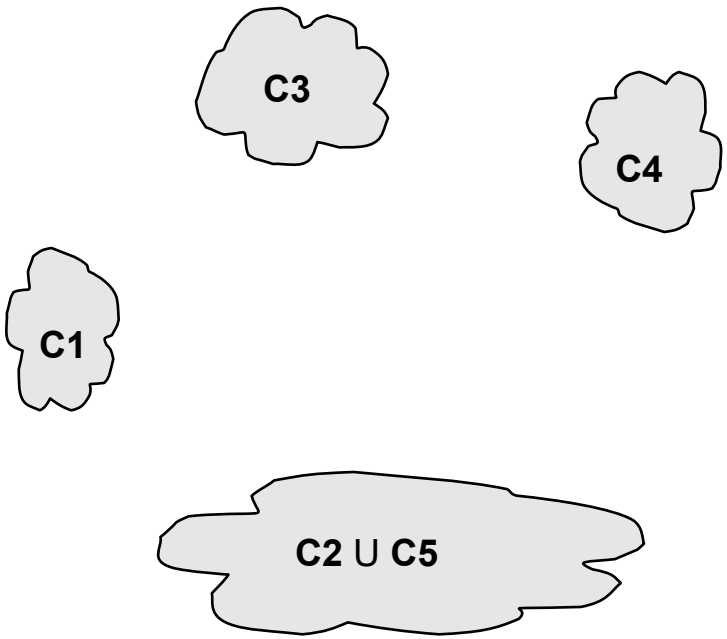
Proximity Matrix



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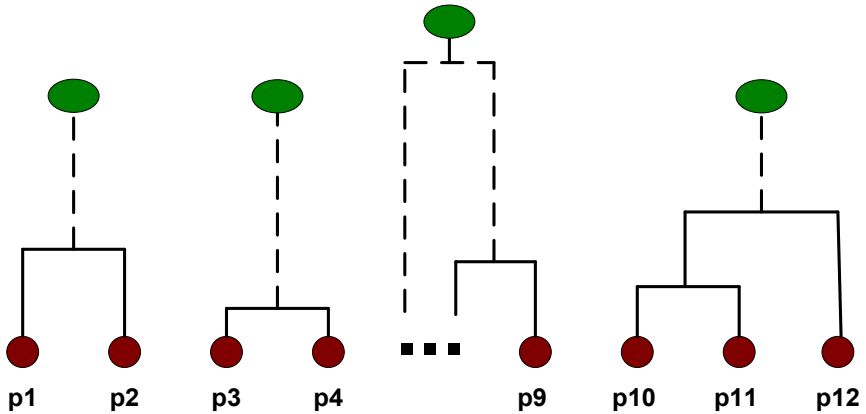
Step 5

- The question is “How do we update the proximity matrix?”



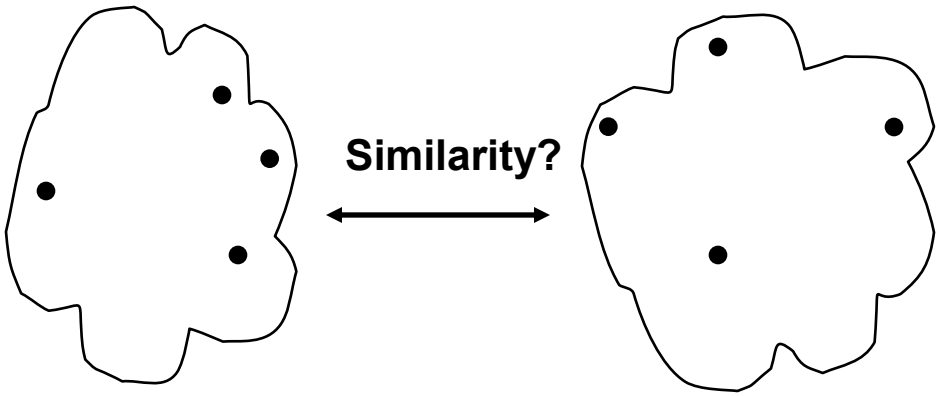
	C1	C2 U C5	C3	C4
C1		?		
C2 U C5	?	?	?	?
C3		?		
C4		?		

Proximity Matrix



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How to Define Inter-Cluster Distance



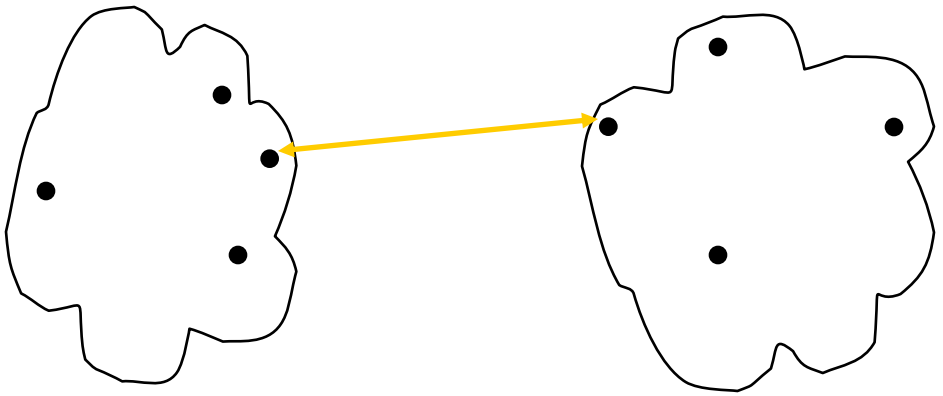
	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						

Proximity Matrix

- MIN
- MAX
- Group Average
- Distance Between Centroids
- Other methods driven by an objective function
  - Ward's Method uses squared error

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How to Define Inter-Cluster Similarity



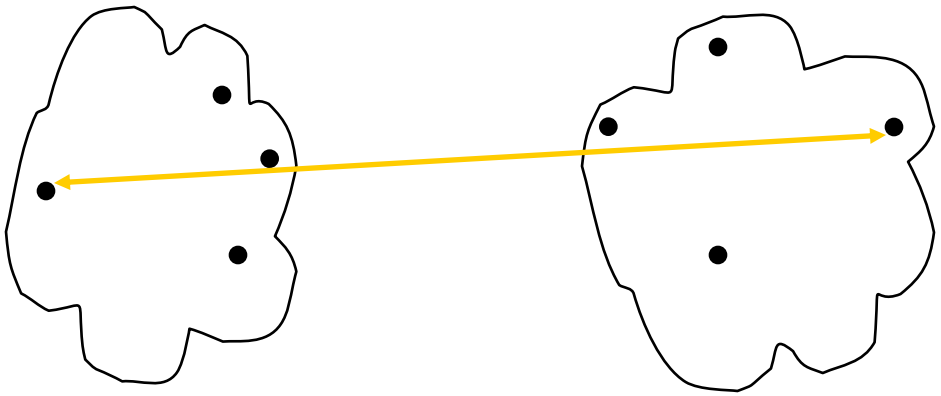
	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

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How to Define Inter-Cluster Similarity



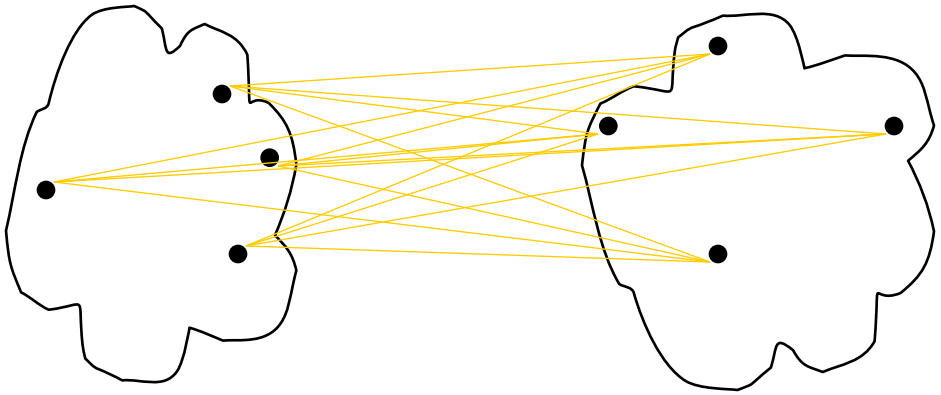
	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

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How to Define Inter-Cluster Similarity



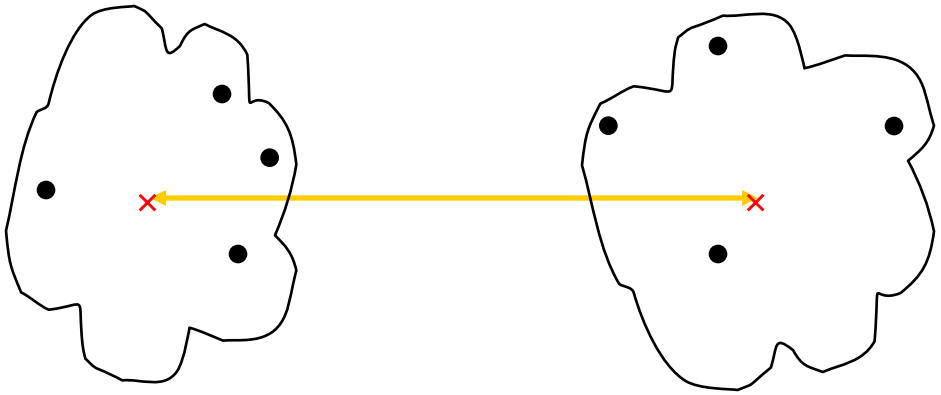
	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						
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.						

Proximity Matrix

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How to Define Inter-Cluster Similarity



	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

- MIN
- MAX
- Group Average
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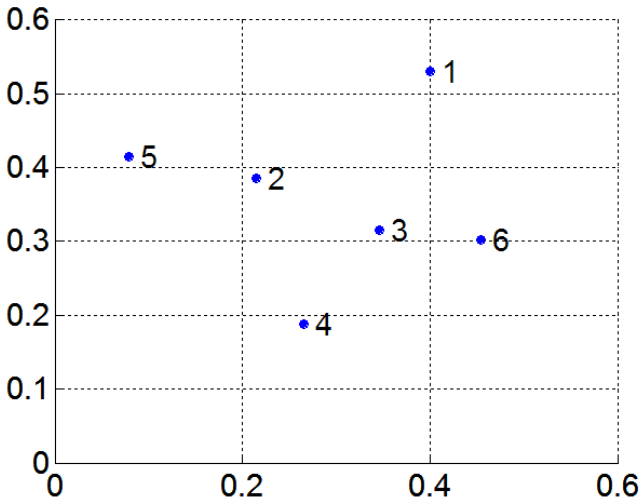
MIN or Single Link

- Proximity of two clusters is based on the two closest points in the different clusters
  - Determined by one pair of points, i.e., by one link in the proximity graph

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Case study #1

Given the distance matrix, use MIN/single link approach for Agglomerative clustering and provide clustering results.



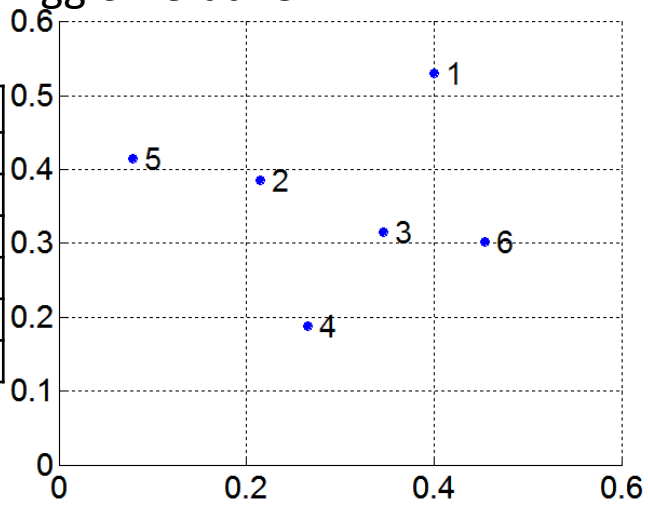
Distance Matrix:

	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

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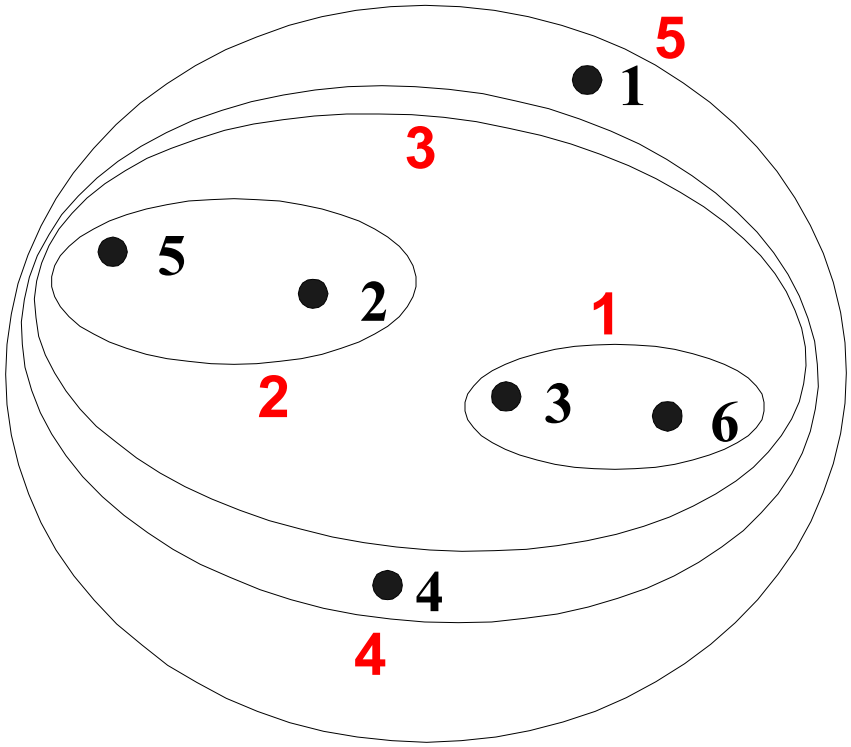
Given the distance matrix, use MIN/single link approach for Agglomerative clustering and provide clustering results.

	p1	p2	p3	p4	p5	p6
p1	0	0.24	0.22	0.37	0.34	0.23
p2	0.24	0	0.15	0.2	0.14	0.25
p3	0.22	0.15	0	0.15	0.28	0.11
p4	0.37	0.2	0.15	0	0.29	0.22
p5	0.34	0.14	0.28	0.29	0	0.39
p6	0.23	0.25	0.11	0.22	0.39	0

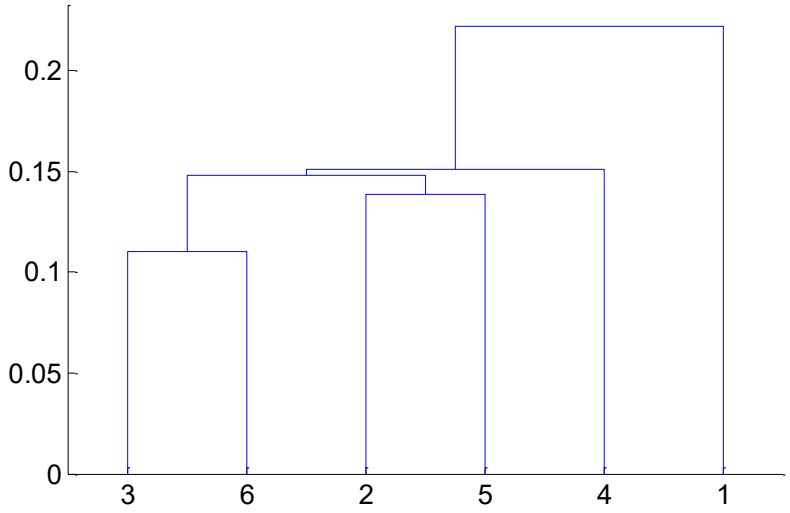


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Hierarchical Clustering: MIN



Nested Clusters



Dendrogram

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Case study #2

Given the similarity matrix, use MIN/single link approach for Agglomerative clustering and provide clustering results.

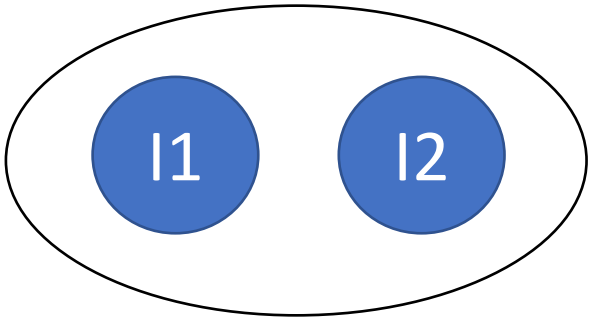
Similarity Matrix

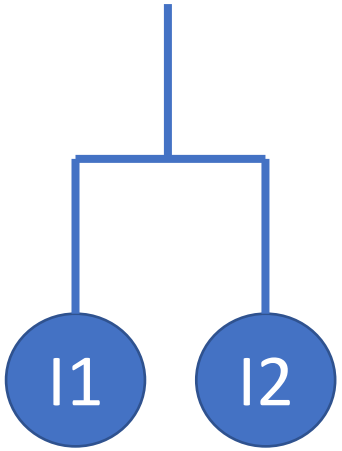
	I1	I2	I3	I4	I5
I1	1	0.9	0.1	0.65	0.2
I2	0.9	1	0.7	0.6	0.5
I3	0.1	0.7	1	0.4	0.3
I4	0.65	0.6	0.4	1	0.8
I5	0.2	0.5	0.3	0.8	1

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Similarity Matrix

	I1	I2	I3	I4	I5
I1	1	0.9	0.1	0.65	0.2
I2	0.9	1	0.7	0.6	0.5
I3	0.1	0.7	1	0.4	0.3
I4	0.65	0.6	0.4	1	0.8
I5	0.2	0.5	0.3	0.8	1





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Similarity Matrix

	I1	I2	I3	I4	I5
I1	1	0.9	0.1	0.65	0.2
I2	0.9	1	0.7	0.6	0.5
I3	0.1	0.7	1	0.4	0.3
I4	0.65	0.6	0.4	1	0.8
I5	0.2	0.5	0.3	0.8	1



	I1+ I2		I3	I4	I5
I1+ I2	1	0.9	0.1	0.65	0.2
	0.9	1	0.7	0.6	0.5
I3	0.1	0.7	1	0.4	0.3
I4	0.65	0.6	0.4	1	0.8
I5	0.2	0.5	0.3	0.8	1

- Update matrix
- Keep the larger similarity

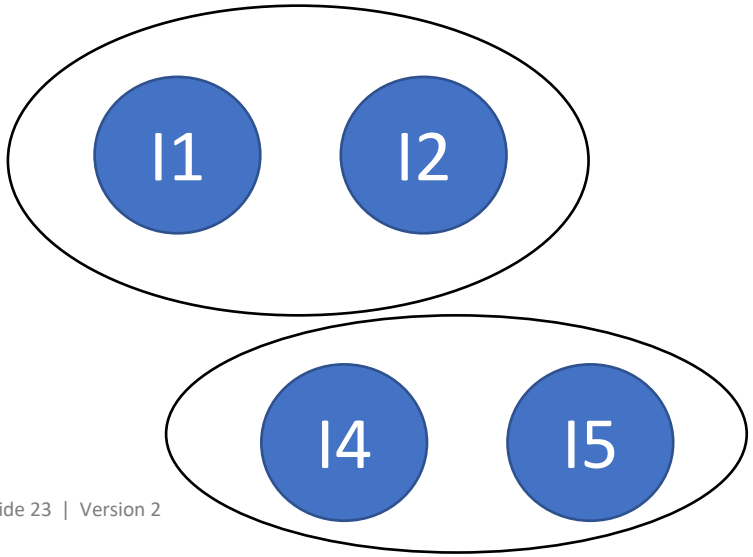


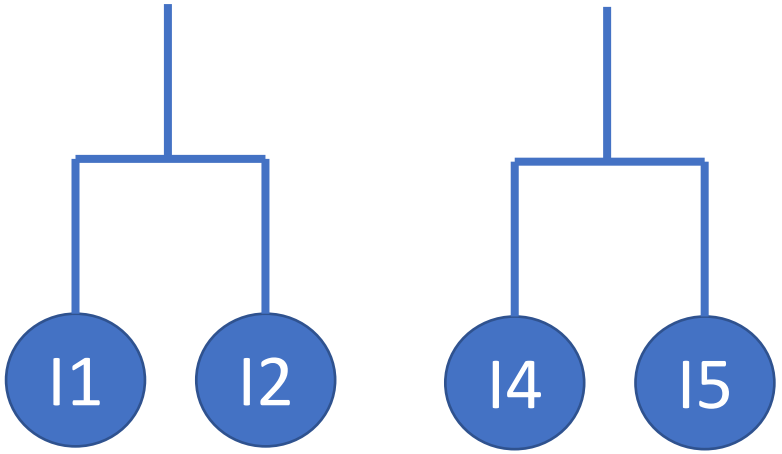
	I1+ I2	I3	I4	I5
I1+ I2	1	0.7	0.65	0.5
I3	0.7	1	0.4	0.3
I4	0.65	0.4	1	0.8
I5	0.5	0.3	0.8	1

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	I1+ I2	I3	I4	I5
I1+ I2	1	0.7	0.65	0.5
I3	0.7	1	0.4	0.3
I4	0.65	0.4	1	0.8
I5	0.5	0.3	0.8	1

	I1+ I2	I3	I4 + I5	
I1+ I2	1	0.7	0.65	0.5
I3	0.7	1	0.4	0.3
I4 + I5	0.65	0.4	1	0.8
	0.5	0.3	0.8	1





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	I1+ I2	I3	I4	I5
I1+ I2	1	0.7	0.65	0.5
I3	0.7	1	0.4	0.3
I4	0.65	0.4	1	0.8
I5	0.5	0.3	0.8	1



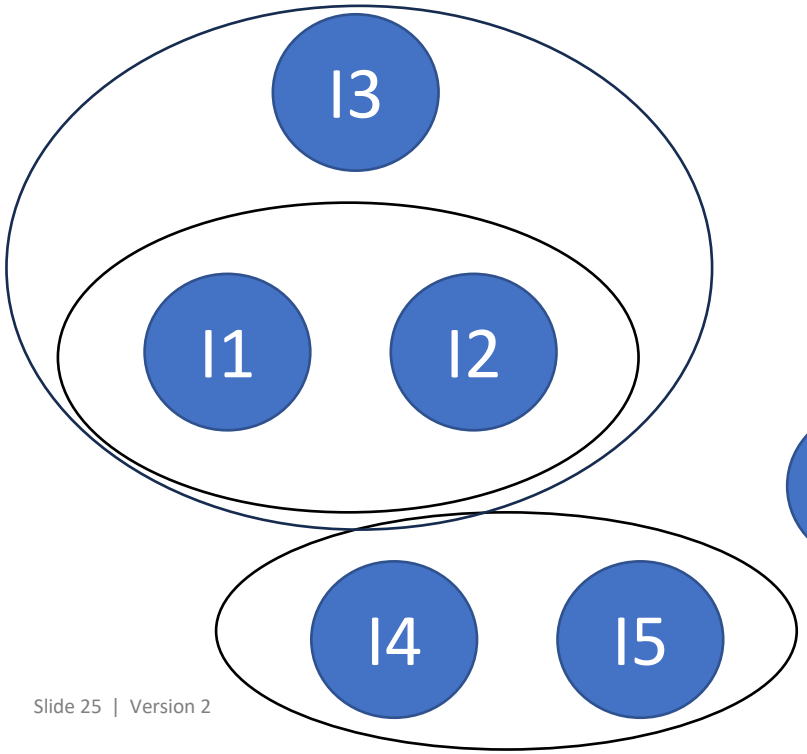
	I1+ I2	I3	I4 + I5	
I1+ I2	1	0.7	0.65	0.5
I3	0.7	1	0.4	0.3
I4 + I5	0.65	0.4	1	0.8
	0.5	0.3	0.8	1

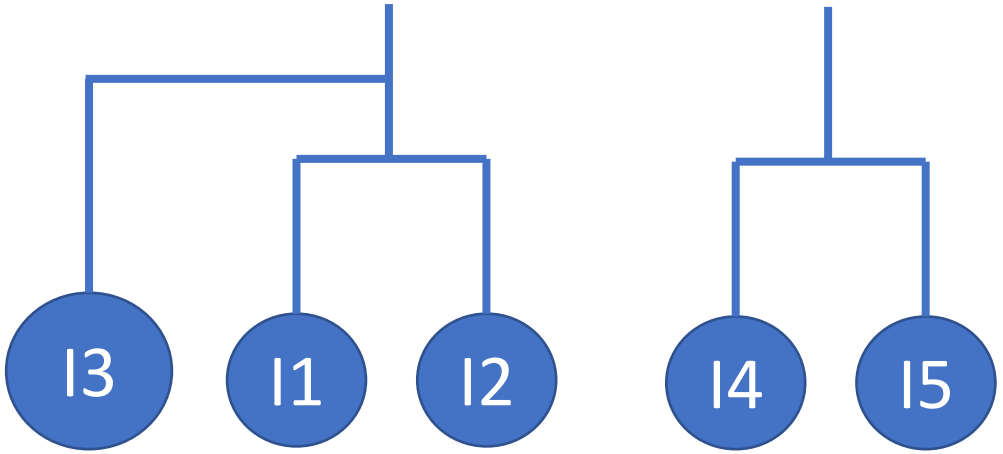
- Update matrix
- Keep the larger similarity

	I1+ I2	I3	I4 + I5
I1+ I2	1	0.7	0.65
I3	0.7	1	0.4
I4 + I5	0.65	0.4	1

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	I1+ I2	I3	I4 + I5
I1+ I2	1	0.7	0.65
I3	0.7	1	0.4
I4 + I5	0.65	0.4	1

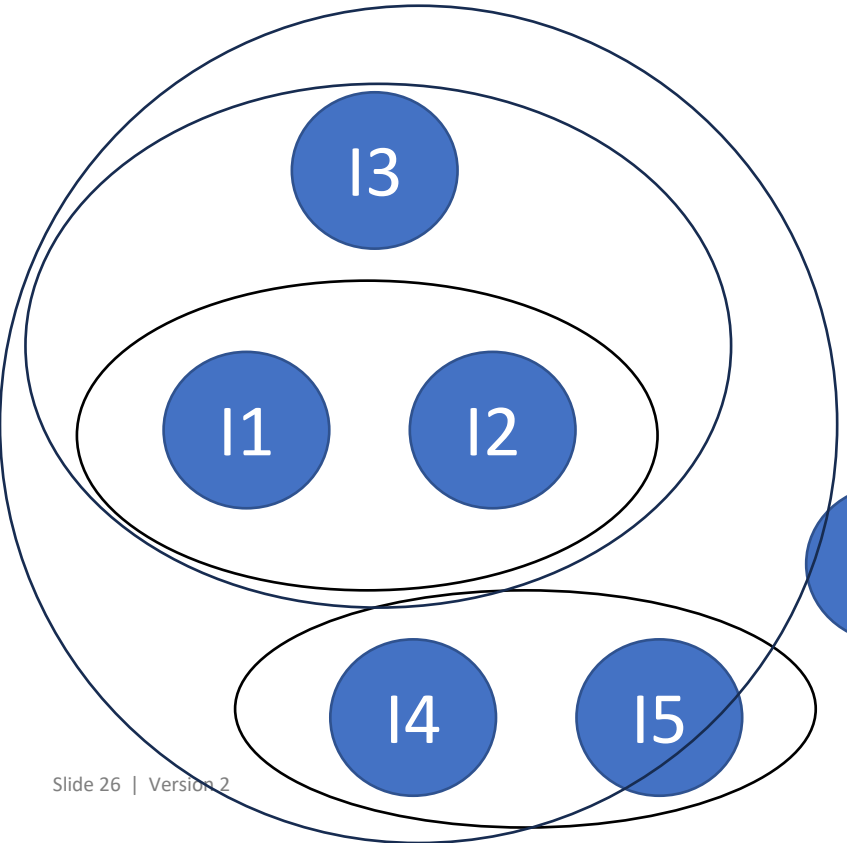


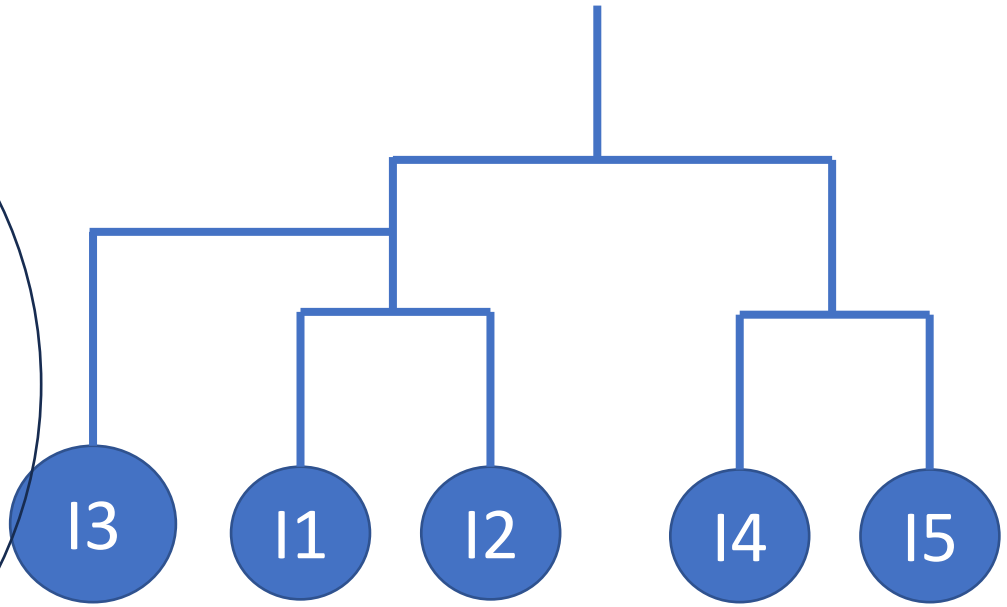


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	I1+ I2	I3	I4 + I5
I1+ I2	1	0.7	0.65
I3	0.7	1	0.4
I4 + I5	0.65	0.4	1

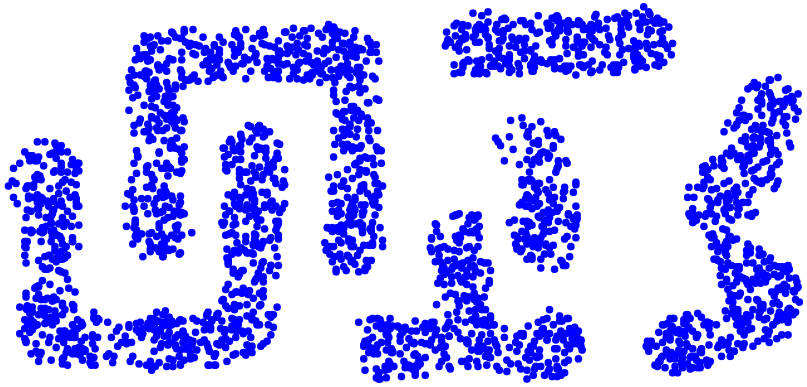
	I1+ I2+ I3	I4 + I5
I1+ I2+ I3	1	0.65
I4 + I5	0.65	1



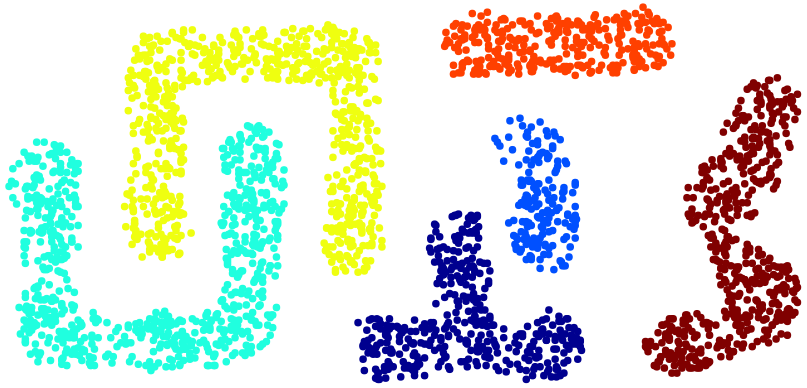


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Strength of MIN



Original Points

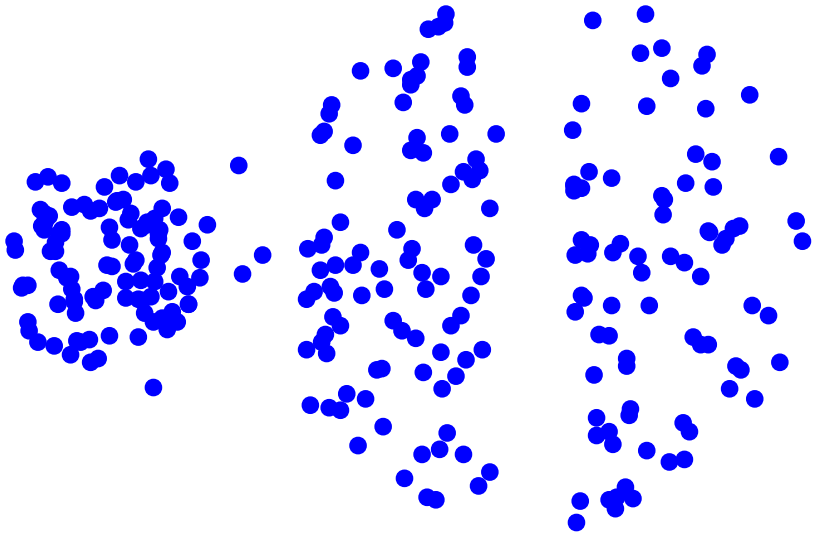


Six Clusters

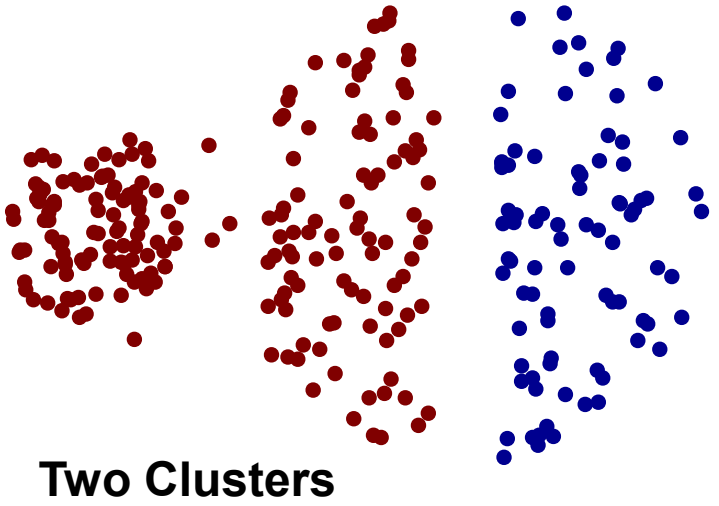
- Can handle non-elliptical shapes

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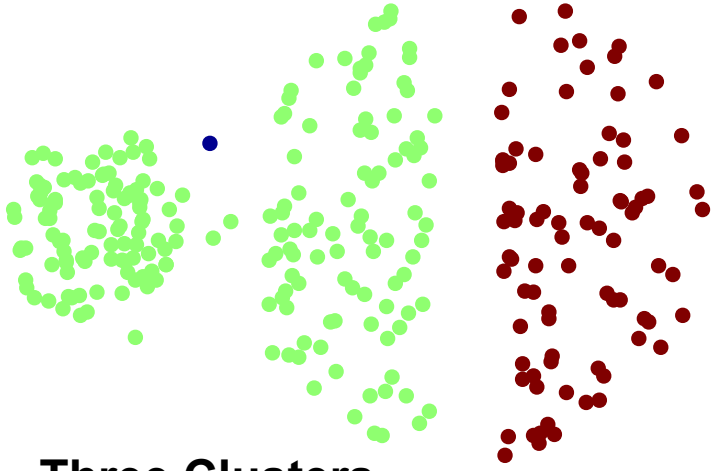
Limitations of MIN



Original Points



Two Clusters



Three Clusters

- Sensitive to noise

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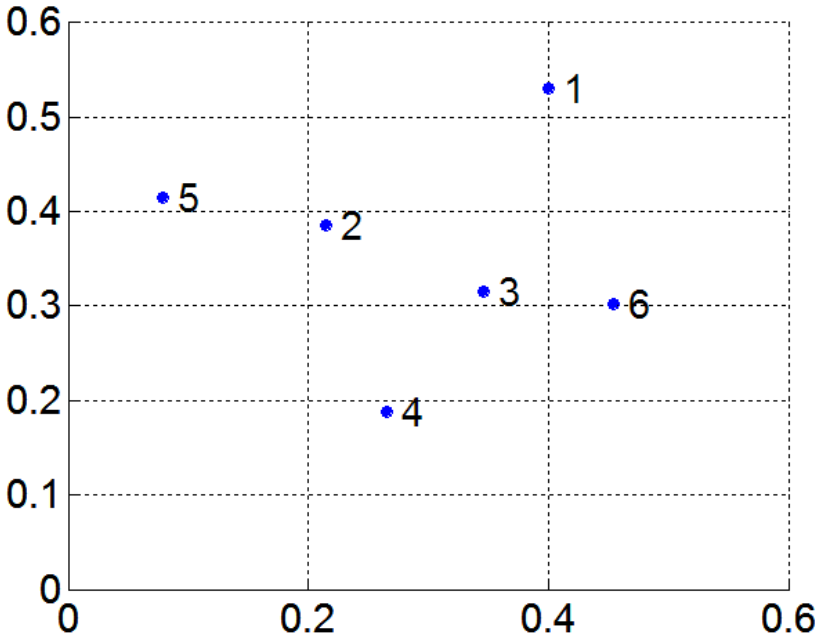
MAX or Complete Linkage

- Proximity of two clusters is based on the two most distant (least similar) points in the different clusters
  - Determined by all pairs of points in the two clusters

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Case study #3

Given the distance matrix, use MAX/COMPLETE link approach for Agglomerative clustering and provide clustering results.



Distance Matrix:

	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

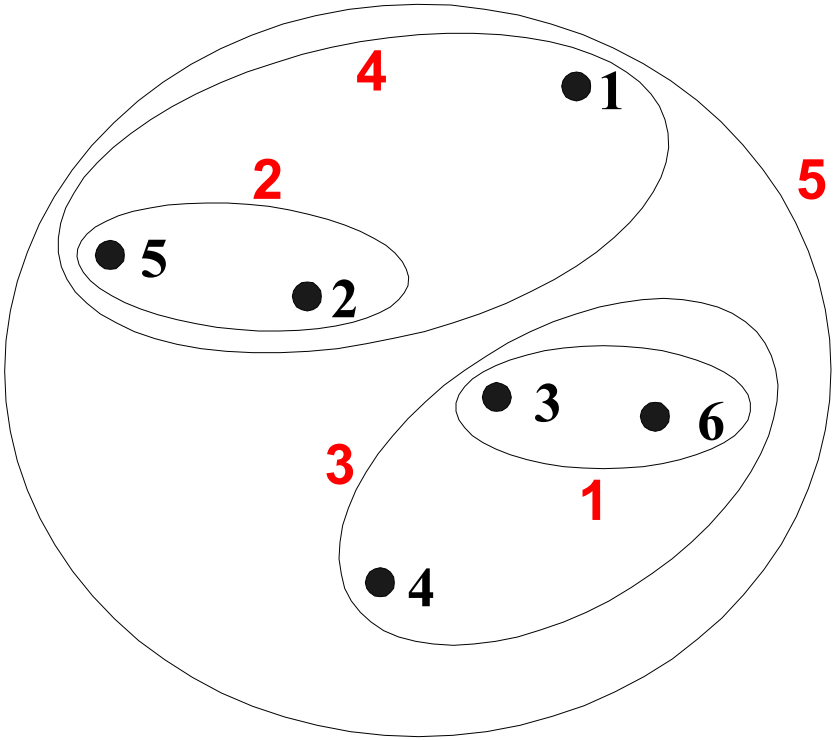
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Given the distance matrix, use MAX/COMPLETE link approach for Agglomerative clustering and provide clustering results.

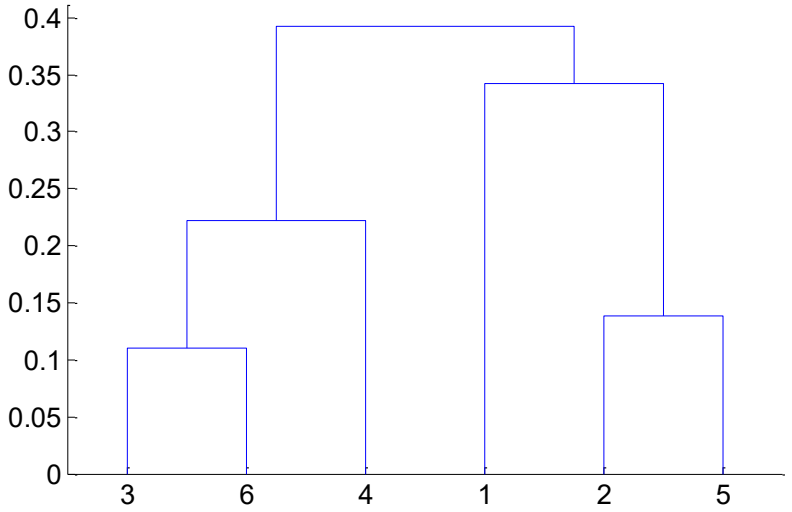
	p1	p2	p3	p4	p5	p6
p1	0	0.24	0.22	0.37	0.34	0.23
p2	0.24	0	0.15	0.2	0.14	0.25
p3	0.22	0.15	0	0.15	0.28	0.11
p4	0.37	0.2	0.15	0	0.29	0.22
p5	0.34	0.14	0.28	0.29	0	0.39
p6	0.23	0.25	0.11	0.22	0.39	0

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Hierarchical Clustering: MAX



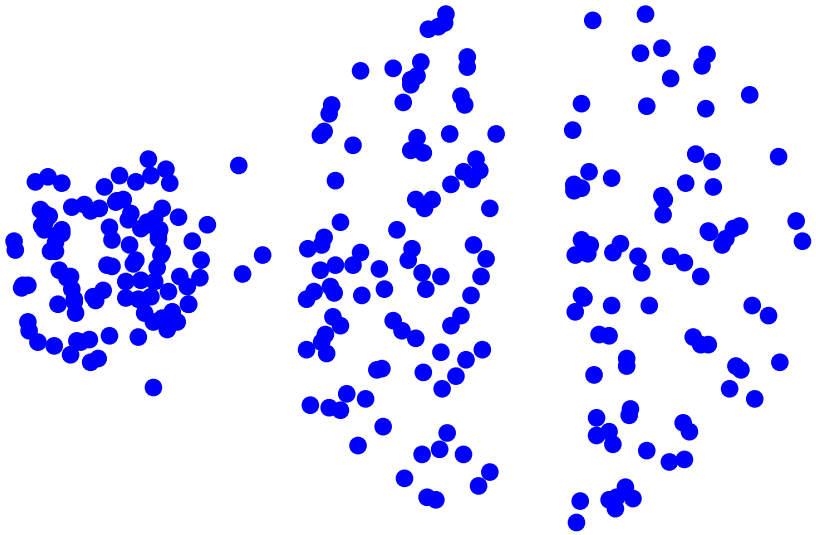
Nested Clusters



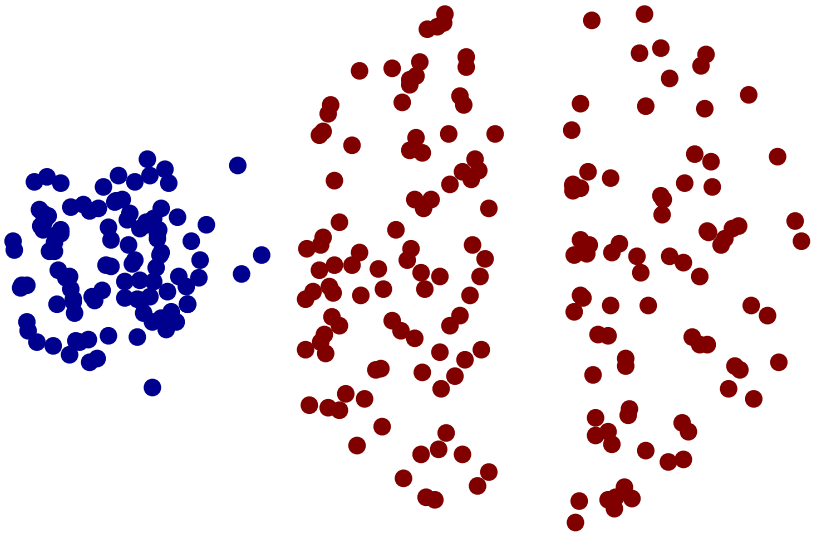
Dendrogram

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Strength of MAX



Original Points

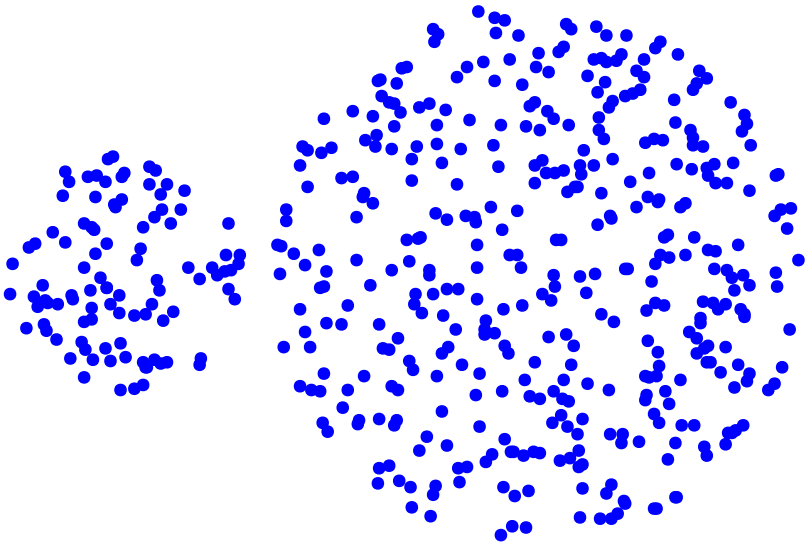


Two Clusters

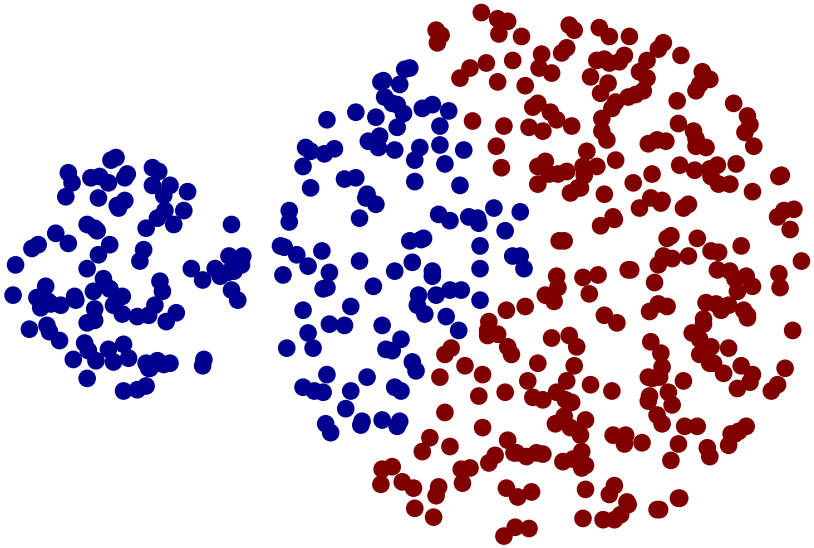
- Less susceptible to noise

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Limitations of MAX



Original Points



Two Clusters

- Tends to break large clusters
- Biased towards globular clusters

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Group Average

- Proximity of two clusters is the average of pairwise proximity between points in the two clusters.

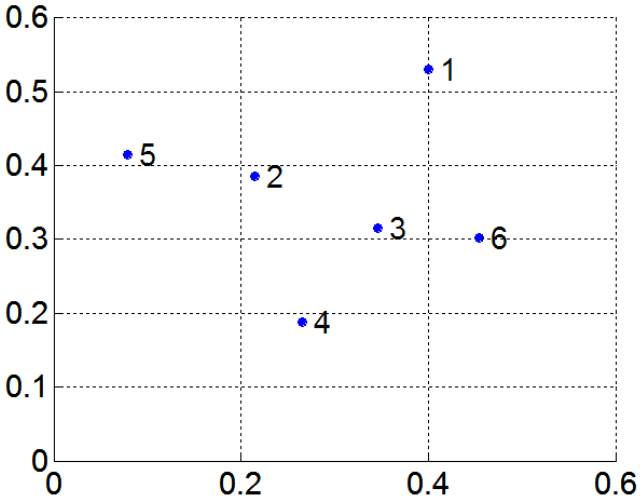
$$\text{proximity}(\text{Cluster}_i, \text{Cluster}_j) = \frac{\sum_{\substack{p_i \in \text{Cluster}_i \\ p_j \in \text{Cluster}_j}} \text{proximity}(p_i, p_j)}{|\text{Cluster}_i| \times |\text{Cluster}_j|}$$

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Group Average

Example:

- dist ({3,6,4},{1}) = (0.22+0.37+0.23) / 3X1 = 0.28
- dist ({2,5},{1}) = (0.24+0.34) / 2X1 = 0.29
- dist ({3,6,4},{2,5}) = (0.15+0.28+0.25+0.39+0.20+0.29) / 3X2 = 0.26



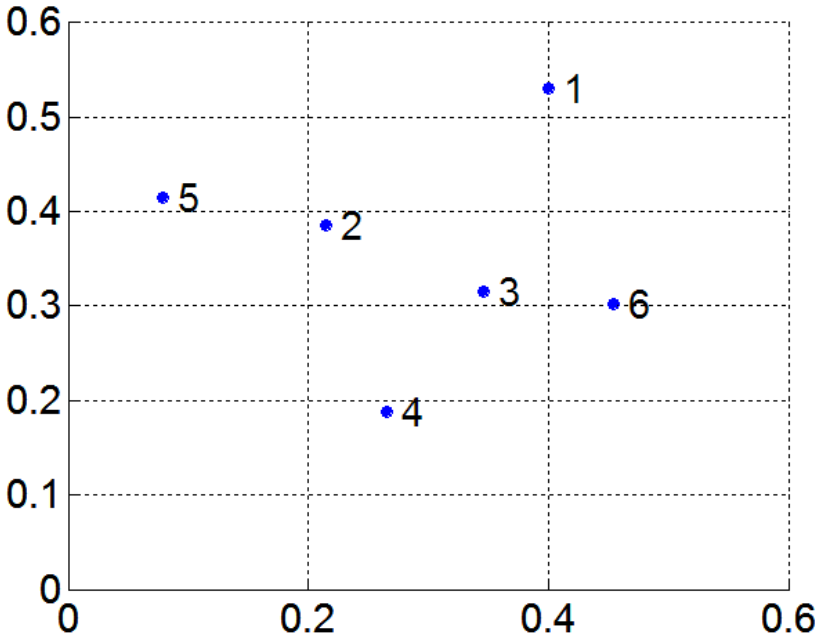
Distance Matrix:

	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

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Case study #4

Given the distance matrix, use group average approach for Agglomerative clustering and provide clustering results.



Distance Matrix:

	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

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Given the distance matrix, use group average approach for Agglomerative clustering and provide clustering results.

	p1	p2	p3	p4	p5	p6
p1	0	0.24	0.22	0.37	0.34	0.23
p2	0.24	0	0.15	0.2	0.14	0.25
p3	0.22	0.15	0	0.15	0.28	0.11
p4	0.37	0.2	0.15	0	0.29	0.22
p5	0.34	0.14	0.28	0.29	0	0.39
p6	0.23	0.25	0.11	0.22	0.39	0



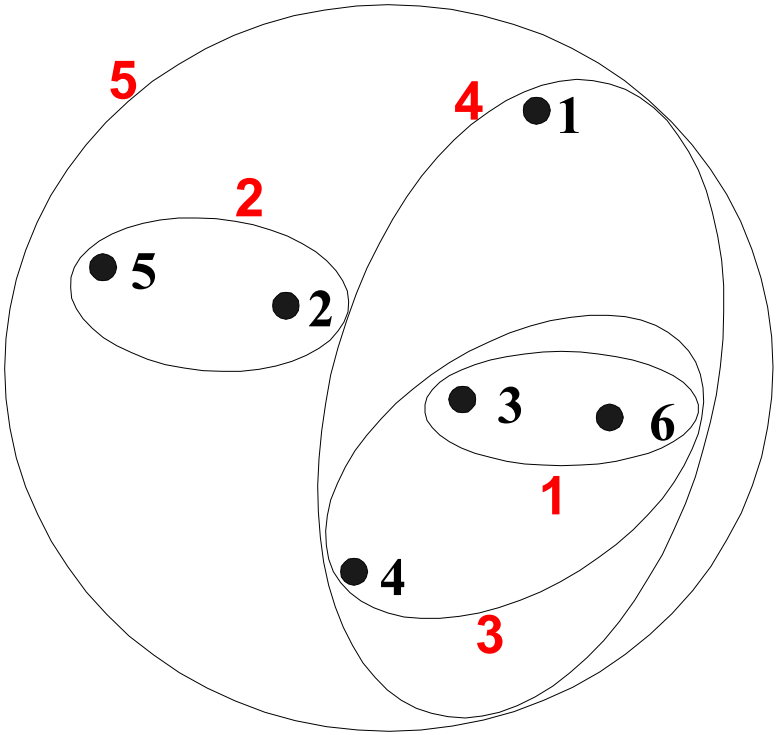
	p1	p2	p4	p5	p6+P3
p1	0	0.24	0.37	0.34	?
p2	0.24	0	0.2	0.14	?
p4	0.37	0.2	0	0.29	?
p5	0.34	0.14	0.29	0	?
p6+P3	?	?	?	?	0

dist({3,6},{1}) = (0.23+0.22) /2 = 0.215

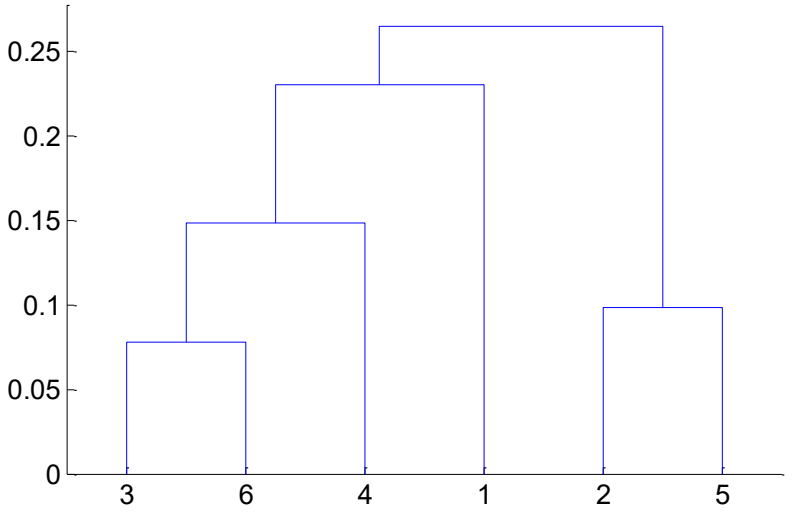
dist({3,6},{2}) = (0.15+0.25) /2 = 0.2

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Hierarchical Clustering: Group Average



Nested Clusters



Dendrogram

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Hierarchical Clustering: Group Average

- Compromise between Single and Complete Link
- Strengths
  - Less susceptible to noise
- Limitations
  - Biased towards globular clusters

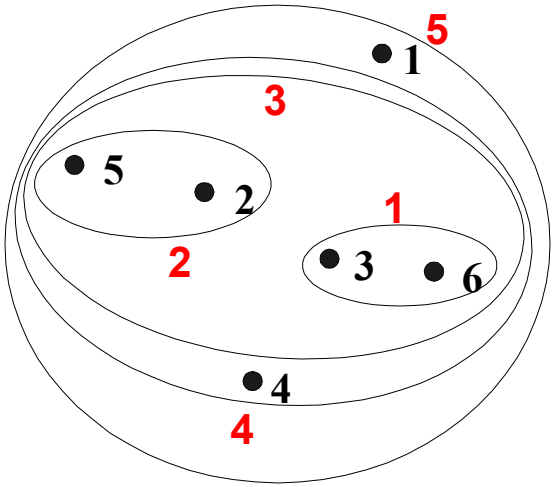
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Cluster Similarity: Ward's Method

- Similarity of two clusters is based on the increase in squared error when two clusters are merged
  - Similar to group average if distance between points is distance squared
- Less susceptible to noise
- Biased towards globular clusters
- Hierarchical analogue of K-means
  - Can be used to initialize K-means

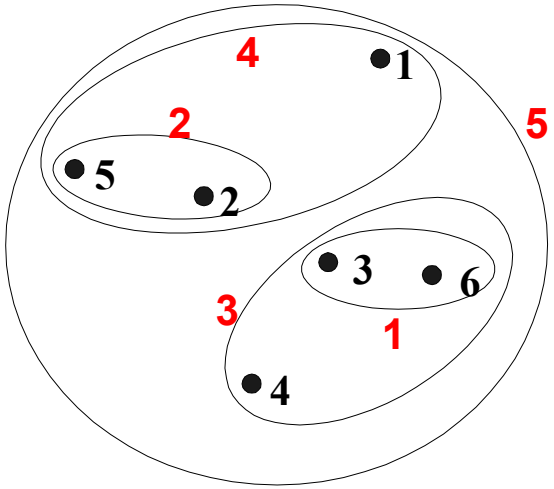
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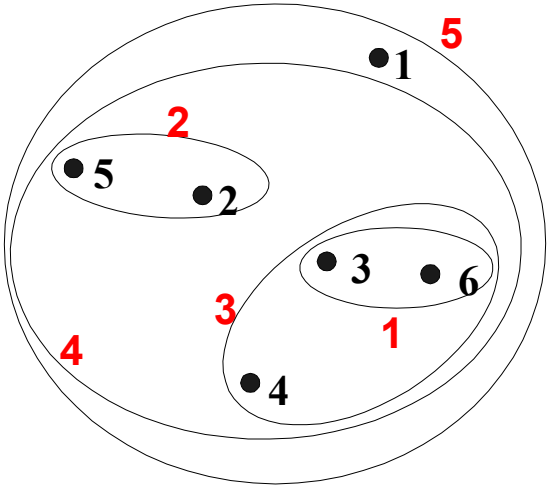
Hierarchical Clustering: Comparison



MIN

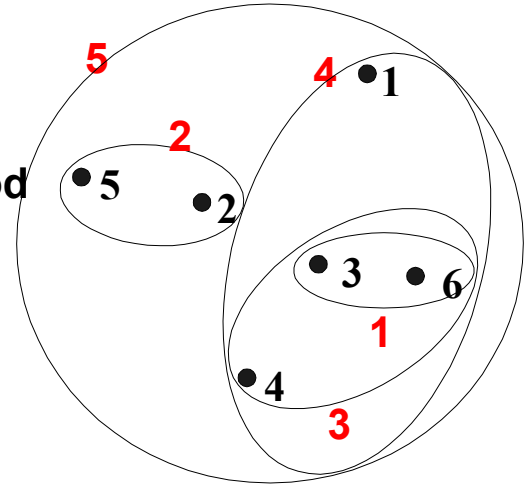
MAX





Group Average

Ward's Method



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Hierarchical Clustering: Problems and Limitations

- Once a decision is made to combine two clusters, it cannot be undone
- No global objective function is directly minimized
- Different schemes have problems with one or more of the following:
  - Sensitivity to noise
  - Difficulty handling clusters of different sizes and non-globular shapes
  - Breaking large clusters

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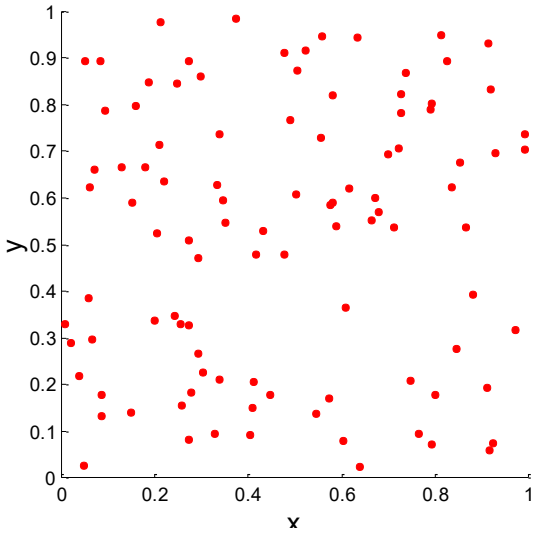
Cluster Validity

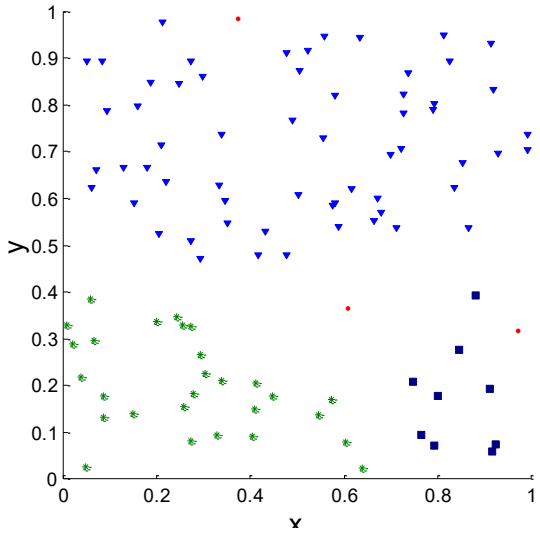
- For supervised classification we have a variety of measures to evaluate how good our model is
  - Accuracy, precision, recall
- For cluster analysis, the analogous question is how to evaluate the “goodness” of the resulting clusters?
- But “clusters are in the eye of the beholder”!
  - In practice the clusters we find are defined by the clustering algorithm
- Then why do we want to evaluate them?
  - To avoid finding patterns in noise
  - To compare clustering algorithms
  - To compare two sets of clusters
  - To compare two clusters

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Clusters found in Random Data

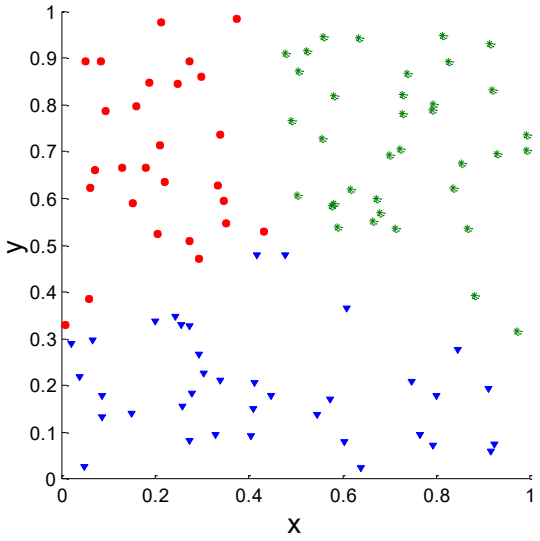
Random Points

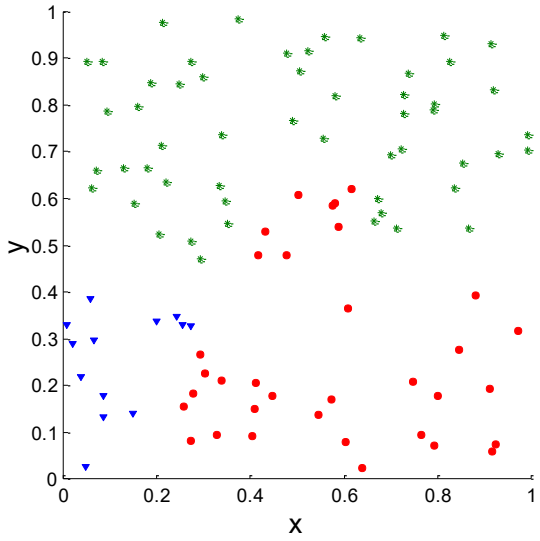




DBSCAN

K-means





Complete Link

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Measures of Cluster Validity

- Numerical measures that are applied to judge various aspects of cluster validity, are classified into the following two types.
  - Supervised: Used to measure the extent to which cluster labels match externally supplied class labels.
    - Entropy
    - Often called *external indices* because they use information external to the data
  - Unsupervised: Used to measure the goodness of a clustering structure *without* respect to external information.
    - Sum of Squared Error (SSE)
    - Often called *internal indices* because they only use information in the data

- You can use supervised or unsupervised measures to compare clusters or clusterings

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Summary

- Hierarchical clustering
- Agglomerative clustering algorithm
- Inter-cluster similarity measurement
- Advantages and challenges

Validation

Next week: Guest lecture

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Thank you

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